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# Blockchain-Based Secure E-Voting DApp

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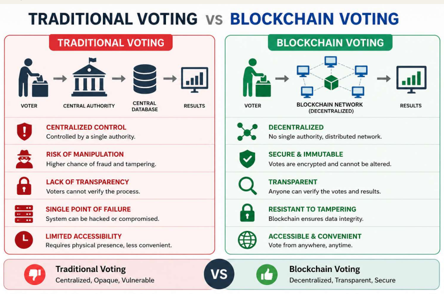
GitHub: https://github.com/danstevemukulupi/BLOCKCHAIN-BASED-SECURED-ELECTRONIC-VOTING-DAPP.git

## Introduction

Traditional voting systems have long been used to elect leaders in democratic societies. However, they suffer from several critical issues, including lack of transparency, risk of fraud, centralized control, and limited voter trust.

This project proposes a Blockchain-Based Secure E-Voting Dapp that leverages decentralized technology to provide a secure, transparent, and tamper-proof voting system. By using blockchain and smart contracts, the system ensures that votes are immutable, verifiable, and anonymous.

The project aims to enhance election integrity, reduce conflicts, and improve accessibility by allowing users to vote remotely using secure digital authentication.




## Objectives

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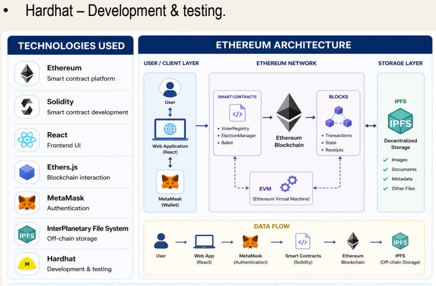
- Develop a secure blockchain-based voting system
- Ensure voter privacy and vote integrity
- Prevent double voting using smart contracts
- Enable transparent and auditable election results
- Integrate wallet-based authentication (MetaMask)
- Store off-chain data using IPFS
- Deploy and test on blockchain testnet


## Technologies Used

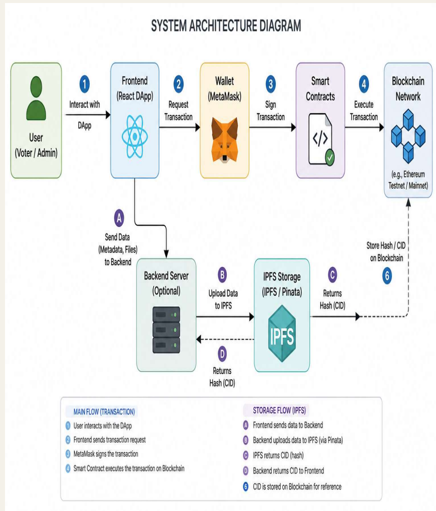
- Ethereum – Smart contract platform
- Solidity – Smart contract development
- React – Frontend User Interface
- Ether.js - Blockchain interaction
- MetaMask – Authentication
- InterPlanetary File System – Off-chain storage
- Hardhat – Development & testing.




## System Architecture

The system consists of multiple integrated components:

- Frontend Interface (React)
- Wallet authentication (MetaMask)
- Smart contracts (Blockchain logic)
- Blockchain network (data storage)
- IPFS (off-chain storage
- Environment setup (Node.js, Hardhat)




## Methodology

Research Methodology

- Studied blockchain and e-voting systems
- Compared traditional vs decentralized voting

Agile Development

- Built system in iterations:
- Registration -> Voting -> Results
- Continuous testing and improvement

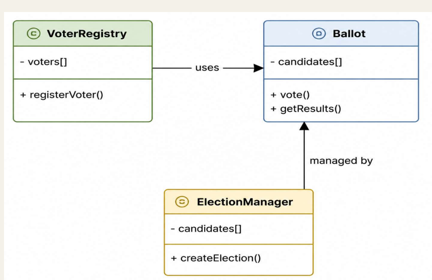
## System Design

Smart Contracts:

- VoterRegistry
- ElectionManager
- Ballot

Key Features:

- Secure vote casting
- Automatic vote counting
- Immutable data storage




## Headers

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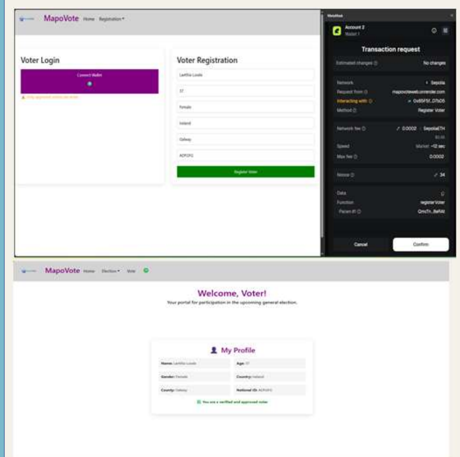
|---|---|
| Introduction, Objectives and Technologies | Conclusion |
| System Architecture, Design and Methodology | References |
| Implementation, Results and Evaluation | Acknowledgements |

## Implementation

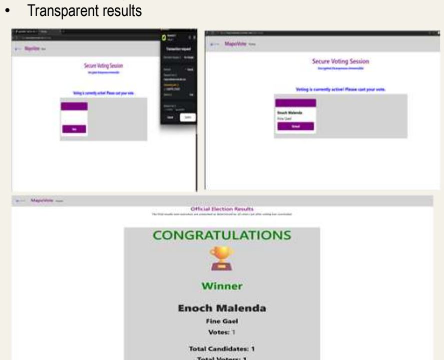
The system was implemented in multiple stages:

- Environment setup (Node.js, Hardhat)
- Smart contract development
- Frontend development (React)
- Blockchain Integration
- IPFS Integration




## Result

- Secure voter registration
- Vote casting through blockchain
- No double voting
- Transparent results




## Evaluation

Strengths:

- High security (immutable blockchain)
- Transparent and auditable
- Prevents double voting
- Decentralized (no central authority)

Limitations:

- Blockchain delays
- Requires technical knowledge

## Conclusion

- This project demonstrates that blockchain technology can significantly improve the security and transparency of voting systems. The developed application provides a reliable and tamper-proof solution that addresses the limitations of traditional voting methods.
- Although challenges such as scalability and usability remain, the system provides a strong foundation for future development in digital voting technologies.

## References

- Blockchain-Based E-Voting Systems: A Technology Review
- Ethereum development documentation | ethereum.org
- IPFS Documentation | IPFS Docs

## Acknowledgements

- Gerard Harisson
- John French
- Atlantic Technological University